

Opto-electrotactile Feedback Enabled Text-line Tracking Control for A Finger-wearable Reading Aid for the Blind

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Abstract—In this paper, to achieve the goal of aiding the blind and visually impaired (BVI) to read any text not written in Braille, a custom-built, finger-wearable, and electro-tactile based Braille reading system with its Rapid Optical Character Recognition (R-OCR) method is developed. The R-OCR is capable of processing text information in real time using a miniature fish-eye imaging device mounted at the finger-wearable system. This allows real-time translation of printed text to electro-Braille along with natural movement of user's fingertip as if reading any Braille sign or book. An electro-tactile neuro-stimulation feedback mechanism is further proposed and incorporated with the reading system, which facilitates a new opto-electrotactile-feedback-based text line tracking control approach that enables text line following by user's fingertip during reading. Extensive experiments were designed and conducted to test the ability of blindfolded participants to read through and follow the printed text lines based on this opto-electrotactile-feedback method. The experimental results show that as the outcome of the opto-electrotactile-feedback, the users who involved in the feedback loop were able to maintain their fingertips within a 2mm distance of the text while "reading" through a printed text line. Our work is a significant step to aid the BVI users with a portable means to read any printed texts to Braille through the following and the translation, whether in the digital realm or physically, on any surface.

I. INTRODUCTION AND RELATED WORK

The number of people of all ages globally that are visually impaired is estimated to be 285 million and is increasing [1]. This is mainly due to increase in screen time usage and also the global trend of aging. To help their reading, the two popular options for this population are either the Braille language or audio means. There are challenges with using the audio as 90% of teachers of blind students agree that technology (such as text-to-audio) should be used only as a *supplement* to Braille rather than as a *replacement* [2]. Also, based on another study, only 10% of blind people who did not know Braille could find a job [3]. These facts show the importance of the Braille in comparison to audio solutions.

The primary benefit of Braille is that it allows users to read in their preferred manner with tactile feeling such as skimming the text, searching for bold or highlighted text and etc. But it also introduces new limitations such as the availability of the Braille books, their size, weight, and cost. As an example, the Webster's Collegiate Dictionary Braille

copy is over 75 volumes, each volume is over three inches thick and costs over five thousand dollars [4].

Currently existing refreshable braille displays are electro-mechanical approaches to translate text to Braille and can solve the problems associated with the traditional Braille systems [5]. However, there are still challenges involved in using a heavy or bulky mechanical device.

Numerous text-to-Braille research has been done in the field including devices such as *Optacon* [6], *OrCam* [7], *FingerReader* [8], *BrailleRing* [9], and *HandSight* [10] where each has proposed different solutions to the problem of converting the written texts to Braille in real time. The use of a camera connected to a regular Optical Character Recognition (OCR) program to extract the text information of the picture is a common practice among all of these approaches. Through electro-mechanical approaches, the output of these devices is given to the user by either mechanically haptic feedback or audio forms.

Our research has introduced a new approach by developing an electro-tactile display that can be used as a portable and refreshable Braille display, in which electrode array of contacts can deliver electrical Braille patterns/signals to the skin without using additional mechanical actuators or parts [11]. An electro-tactile stimulation based display has numerous advantages over the the mechanical approaches. These include being significantly compact, being easier to fabricate, and also having lower power consumption. The absence of any mechanical parts will lead to ease of repair and lower production costs as well. The main concepts and mechanism of an electro-tactile display have been investigated mostly by Kaczmarek and Tyler [12][13]. For our electro-tactile display, a new adaptive spatial mapping strategy was developed to supply the appropriate stimulation feeling [14]. Different from their approaches, our approach is to put the active electrode array slightly distal to the fingertip vortex. A bigger single return electrode is then placed on the palm of the hand as the return electrode. This configuration can offer the advantages of allowing more active electrodes to fingertip so as to best match the density of the tactile receptors in situ [15][16].

Based on our electro-tactile display array configuration and methods, this work focuses on the feedback that the user needs to perceive so as to maintain their fingertip on the text line while "scanning" the printed texts. The feedback mechanism measures the appropriate distance to the line and gives the corresponding electro-tactile stimulation feedback to the user as to adjust the fingertip up or down along the text line. These feedbacks are simultaneously given through

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our electro-tactile display array on user's fingertip. A preliminary investigation of users recognition of the directional commands was done in our previous research [17]. It was found that the users had more difficulty in identifying the moving pattern if the moving direction is horizontal rather than vertical. This was further verified by combining the results of the movements by disregarding the effects of the diagonal patterns [17]. Based on these findings, the directional feedback in this work is more related to vertical movements. Our approach is verified that it helps keep the users' fingertip within 2mm distance of the followed printed text line, which ensures providing an exceptional accuracy in tracking the text line for the BVI users.

The contribution and novelty of this work lie in 1) development of a Rapid Optical Character Recognition (R-OCR) method for real-time processing text information based on a fish-eye imaging device mounted at the finger-wearable electro-tactile display. This allows real-time translation of printed text to electro-Braille along with natural movement of user's fingertip as if reading any Braille sign or book; 2) proposing an electro-tactile neuro-stimulation feedback mechanism and incorporating it with the R-OCR method, to further facilitate a new opto-electrotactile-feedback based text line tracking control approach, so as to enable text line following by user's fingertip during "reading"; 3) extensive experiments were designed and conducted to test the performance of the methods and systems as well as validating the ability of blindfolded participants in using the system. Our work is the first-ever synergetic effort to tackle the challenges from three angles (i.e., perception, computer vision, and human-device interaction) that will produce a holistic and intelligent solution for a complete reading aid for BVI.

This paper is structured into the following four sections: The aid system and methods; Experimental procedure and setup; Results and discussions; and Conclusion.

II. THE WEARABLE AID SYSTEM AND METHODS

A. Hardware and System

The interactions of the users with the system are through a finger-wearable Braille display. The developed Braille display is based on an 8-dot Braille system similar to the Gardner-Salinas Braille codes commonly known as GS8 [18]. This 8-dot system has two extra dots on the bottom row. This gives an added benefit of being able to display special characters in mathematics and science such as Greek letters, mathematical symbols and also Unicode characters. All these added benefits are alongside supporting the traditional 6-dot Braille.

We extended this 8-dot system to a 16-dot display by adding two more columns on each side. We use these extra 8 dots for sending commands required for controlling the movements of the fingertip on the text line.

A single board computer (SBC) was used for the purpose of portability of the system. The SBC used in this research was chosen as a Raspberry Pi 3 (RPi3) with 1GB RAM and a CPU frequency of 1.2GHz. The camera used in this research is a tiny I2C fish-eye camera with CSI conversion.

The wearable aid system and its components are shown in Fig. 1. The developed electro-tactile based Braille display can also be seen in this figure. The display supplies the fingertip with a stimulation voltage necessary to produce almost 30μA.

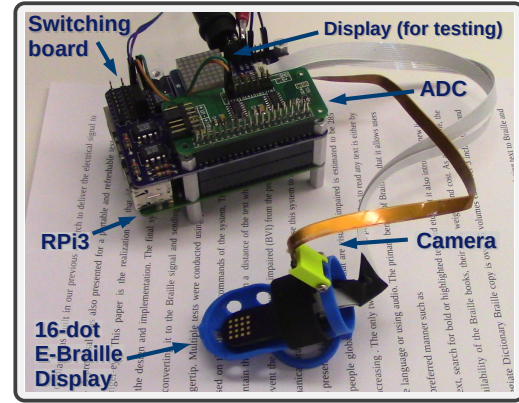


Fig. 1. The wearable aid and its components. Note that the RPi3 can work as a stand-alone system without any dependency on a PC. The ADC and the switching board are to control the voltage and current passing through the skin.

B. The Rapid OCR for Fish-Eye Camera

Fig 2 shows the view field of the fish-eye camera when the wearable system touches on the text page. The text images captured by the camera will then be processed through a proposed Rapid OCR (R-OCR) method. Our preliminary results demonstrate the proposed R-OCR method can achieve average accuracy of 97.5% but with low (highly competitive) computational time [11]. There are multiple technical steps and challenges for the image capture using the camera and for incorporating with the R-OCR method. When the camera captures images from non-planar objects, there are usually distortions in the texture image of curved surface. For better performance of image recognition by the OCR, a transformation needs to be applied to flatten such images. The common image transformation such as perspective and affine transformation can work correctly only for planar images. To solve this issue, here we first employ the thin plate splines (TPS) approach as the non-rigid transformation model. Suppose x is two dimensional, the TPS fits a mapping function $f(x)$ between corresponding point-sets y_i and x_i that minimizes the following energy function. The relationship is revealed in (1).

$$E_{tps}(f) = \sum_{i=1}^K \|y_i - f(x_i)\|^2 \quad (1)$$

In this approach, the smoothing variant uses a tuning parameter λ to control how non-rigid is allowed for the deformation, which balances the aforementioned criterion with the measure of goodness of fit. Thus, the minimization becomes:

$$E_{tps,smooth}(f) = \sum_{i=1}^K \|y_i - f(x_i)\|^2 + \lambda \int \int [(\frac{\partial^2 f}{\partial x_1^2})^2 + 2(\frac{\partial^2 f}{\partial x_1 \partial x_2})^2 + (\frac{\partial^2 f}{\partial x_2^2})^2] dx_1 dx_2 \quad (2)$$

Following (2), results are achieved in image flattening from non-planar surface (e.g. cylindrical surface) to planar surface. Another challenge is the blurring or degradation of an image which can be caused by the movement of the camera during the image capture process or out-of-focus optics. A blurred image can be approximately described by:

$$y = h \otimes x + n \quad (3)$$

here y is the blurred image, x is the original image, n is the additive noise that is introduced during image acquisition, h is the point spread function (PSF), and $\otimes$ is the convolution operator. The blurring is typically modeled as the convolution of a PSF with a hypothetical sharp input image, where both the sharp input image (which is to be recovered) and the PSF are unknown. In almost all cases, there is insufficient information in the blurred image to uniquely determine a plausible original image.

Our solution for de-blurring considering (3), is blind deconvolution that allows the recovery of the target scene from a blurred image in the presence of a poorly determined or unknown PSF. The goal of blind deconvolution is to infer both f and h given a single input g . The approach here is a maximum-a-posterior (MAP) estimation, seeking a pair $(\hat{x}, \hat{h})$ minimizing:

$$(\hat{x}, \hat{h}) = \underset{x, h}{\operatorname{argmin}} \lambda \|x \otimes h - g\|^2 + \sum_i |g_{x,i}(x)|^a + |g_{y,i}(x)|^a \quad (4)$$

By solving (4), a de-blurred image is obtained through the blind deconvolution approach.

Once we achieve good-quality images from text surfaces, the next step is to check the orientation of the text document. The document might have been slightly rotated during acquiring images. Detecting the skew angle and rotating the text to be up-right before the subsequent procedures will be a necessary step for our approach. The popular technique for text line detection and then skew angle is the Standard Hough Transform (SHT). However, performing SHT is time consuming. As a solution, Probabilistic Hough Transform (PHT) is selected as it does not consider all the points, instead, it takes only a random subset of points that is sufficient for our text line and skew angle detection subroutine.

C. Detection and Opto-electrotactile Feedback

When the BVI user wears the finger-wearable device and moves the finger to scans texts on pages, multiple text lines can be captured within the camera's field of view. It is therefore needed to identify which word is the center of focus and is going to be recognized. To be convenient, the movable fingertip of BVI user can be set as a reference for finding words. Therefore, real-time detecting the fingertip in the image and acquiring the position of the fingertip relative to the text lines become necessary. The detection and feedback scheme is then developed for guiding the BVI users to move their fingertip to follow text lines on the page, so as to efficiently "read" texts. The diagram of the scheme

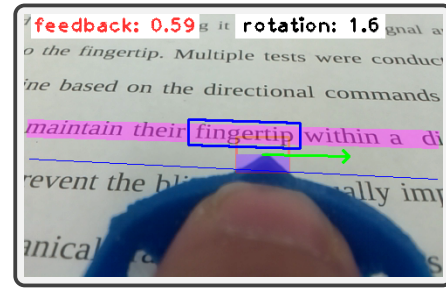


Fig. 2. The camera view and text lines. The distance from the line is shown as the "feedback" parameter and the angle is shown with the "rotation" parameter. The image processing algorithm has found the word "fingertip" in this picture and it was shown in a box. This boxed section is later corrected and sent to the OCR.

is shown in Fig. 3 and it consists of four technical methods: fingertip detection, text line detection, word recognition, and the feedback for guiding finger movement, as detailed below.

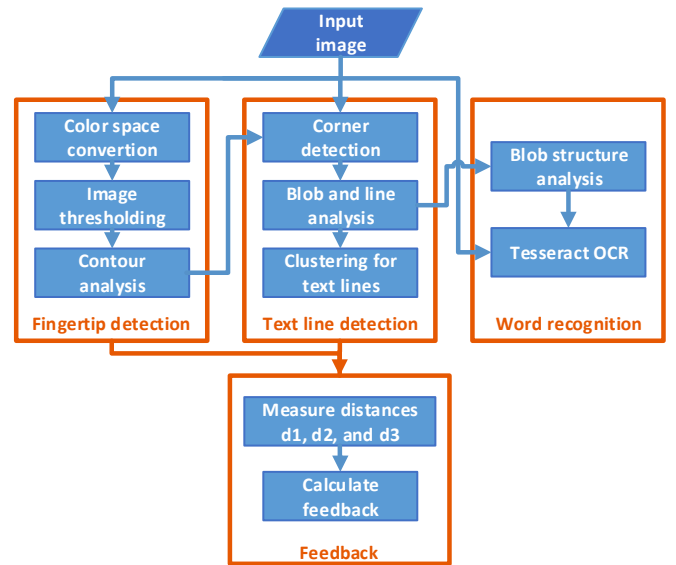


Fig. 3. The diagram of the detection and feedback scheme.

1) *Fingertip Detection*: As shown in Fig. 2, the blue color of the finger-wearable device helps make the device different from the page background in the image. To detect the fingertip (i.e. the sharp tip of the blue device), we first convert the RGB input image to the $L^*a^*b^*$ color space. The chromaticity-layer 'b*' indicates where the color falls along the blue-yellow axis. Because the blue color can be visually distinguished from others in the b^* channel, this allows us to use the Otsu's method that performs clustering-based image thresholding automatically to segment the fingertip from the background. The fingertip (the sharp tip) can then be detected as the topmost pixel on the contour of the largest object. Fig. 4 shows the intermediate and final results of the fingertip detection.

2) *Text Line Detection*: Instead of using adaptive thresholding in a sliding window for text line detection [8][19], we apply the FAST algorithm [20] to help detect corners rapidly

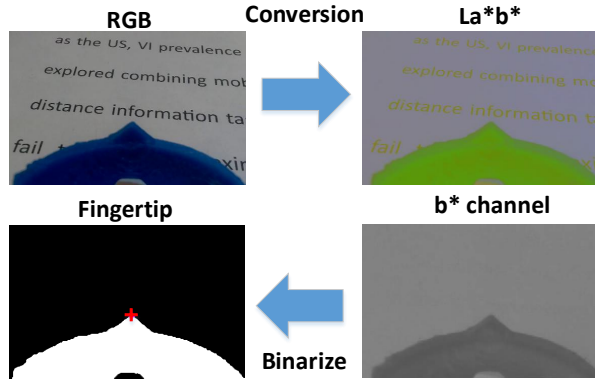


Fig. 4. The fingertip detection process.

in images. The binary mask generated in the previous stage is then used to remove corners that are not on the page surface. As a result, the left corners in image are only from text lines. The FAST line detection algorithm [21] is then applied to find the line segments in the binary image. The angle of the longest line segment is selected as the skew angle of whole text block. Eventually, these rotated corners are clustered to several groups according to their y-coordinate values. The group of corners are transformed inversely to original image to form the regions of text lines. Fig. 5 shows a process of the text line detection.

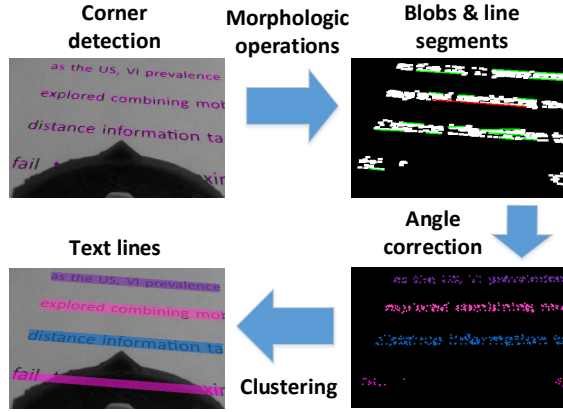


Fig. 5. The text line detection process.

3) *Word Recognition*: Based on acquired positions of the fingertip and the text lines, word recognition in the zone of positions will be conducted. The first step is to extract the focused word from the text line. The blob image obtained in the previous stage is reloaded and the Connected Component Analysis is employed for finding the largest blob in the text line above and closest to the fingertip. We then search for other blobs whose bounding box are overlapping with the largest blob, and finally merge them to form the word blob. The next step is to crop the word region from the original gray image according to the bounding box of the word blob. At the end, the R-OCR that is based on a Tesseract OCR engine [22] is loaded for rapid word recognition. Fig. 6 shows an illustration of the word extraction.

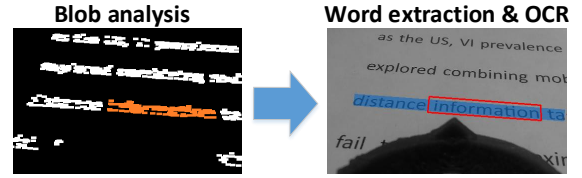


Fig. 6. The word extraction and recognition process.

4) *Opto-electrotactile Feedback Guided Finger Movement*: The motion estimation of the fingertip on the page surface is useful for the analysis of the fingertip movement and for the feedback that controls fingertip's movement on the page. Relying on the visual information we achieved from the camera, the electrotactile stimulation feedback that guides the BVI users to trace their fingers over the text lines can be realized. In this method, we select a baseline to equally split the area between the two text lines from the optical image captured by the camera. Three distances are then defined as the stimulation feedback parameters shown in Fig. 7: d_1 is the distance between the baseline and the upper text line, d_2 is the distance between the baseline and the lower text line, and d_3 is the distance between the baseline and the fingertip.

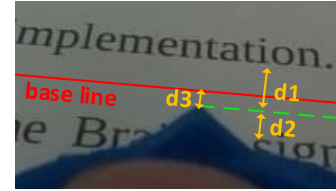


Fig. 7. The stimulation feedback parameters based on three distances from optical image.

Based on three distances, we can define the electrotactile stimulation feedback strength factor, s , as follows:

$$s = (-1)^k \left(\frac{d_3}{d_3 + \min(d_1, d_2)} \right), \quad (5)$$

$$k = \begin{cases} 0, & \text{if fingertip is above baseline} \\ 1, & \text{otherwise} \end{cases}$$

The sign of s indicates the direction of movement of the fingertip. For instance, if the sign is positive, this means the user is moving their fingertip below the baseline. If negative, this indicates moving fingertip above the baseline. The absolute value of s , which is in the range of $[0,1]$, reveals the needed strength of stimulation signal for electrotactile-based Braille display to pull the finger back to the baseline. Fig. 8 demonstrates two cases where a fingertip is moving above and below the baseline with the stimulation feedback factor calculated based on three distances.

By using the feedback factor defined in (5) and the triggered electro-tactile stimulation from the display, the BVI users are in the tracking control loop and will be able to move their finger around the baseline following the stimulation guidance. In addition, by computing the distance between the

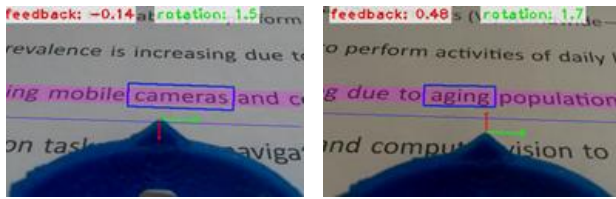


Fig. 8. The two cases for finger moving up and down with the feedback factors. The feedback factor triggers the electro-tactile stimulation through the display.

fingertip and the endpoints of a text line, we can determine if the fingertip is located near the beginning of the text line or is approaching the end of the text line. Based on current system performance, two to three frames of video per second can be processed.

III. EXPERIMENTAL PROCEDURES AND SETUP

A. Participants

There were six subjects participated in the experiments. They were all healthy graduate students and researchers at the University of Nevada, Reno. At this stage, all participants were sighted but were blindfolded for the purpose of the experiments (Note that the blind volunteers were temporarily unavailable due to the COVID pandemic.). The future experiments will include recruiting BVI participants for comparison. Participation was with informed consent and followed protocols approved by the University of Nevada, Reno Institutional Review Board.

B. Experimental Procedure

Two experiments were designed for comparing the effectiveness of the proposed opto-electrotactile feedback strategy for text line tracking, as shown in Fig. 9. Note that in the printed text page we chosen for the experiments, each line of the text has a height of 3mm with a distance of 10mm from the lines above or below (center-line to center-line). In

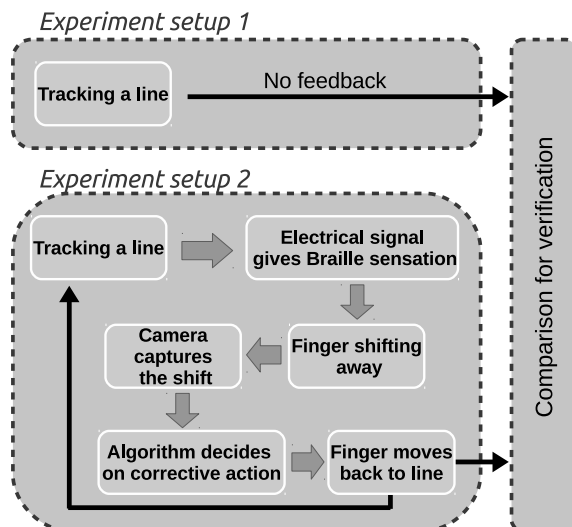


Fig. 9. Two designed experiments and procedures.

the first experiment, a printed text was presented to the user subjects and they were asked to place their fingertips at the beginning of a line. Then by closing their eyes, they were required to move finger to track the line without any feedback. A top-view camera was set up to capture the movement of the subject's fingertip. In the second experiment, by turning on the electro-tactile display system, the moving up and down stimulation commands were introduced to user subjects as an alternating signal with half a second delays. The next step was a test to find out whether they could identify each stimulation signal. In this test, a random command (either moving up or down) was given to the user fingertip and the user was asked to identify the command. The trial was repeated 10 times. Lastly, when users were comfortable in identifying the moving up and down stimulation commands, they were asked to place their fingertips at the beginning of a line and close their eyes and start moving the finger to track the line. As their finger started to drift from the line (either going up or down), the system gave them the appropriate stimulation feedback (either moving up or down) to correct their fingertip position. The process was continued until reaching the end of the line and repeated 3 times for each user. The whole session was captured in video via the top-view camera. The videos were later processed and analyzed to show the tracking path that each user's fingertip traveled on the page with or without the feedback.

IV. RESULTS AND DISCUSSIONS

A. Comparison of Line Tracking With and Without Feedback

As it was described in the above section, a comparison was conducted between tracking the text line by fingertips with and without the opto-electrotactile feedback. It is clear that without the feedback, the tracking almost get to 20mm away from the baseline, while the feedback tracking control method was able to improve the drifting of the fingertip and maintain its movement within 2mm distance of the baseline. Fig. 10 plots the comparison result. Fig. 11 shows the histogram of controlled fingertip location with the feedback. The histogram is achieved from 11440 data points of 25 tests.

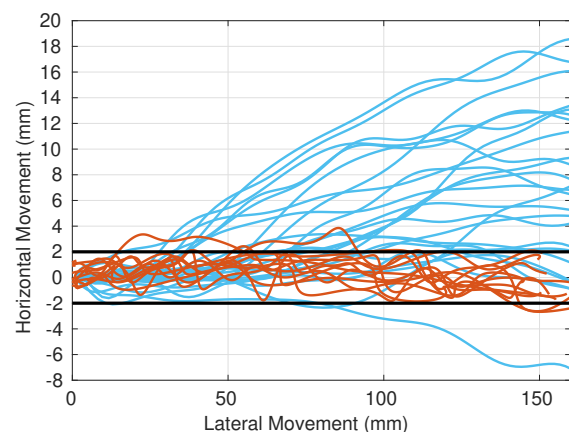


Fig. 10. Comparison result: The blue lines are the natural fingertip movements in the absence of any feedback. The brown lines are the finger movements under the feedback.

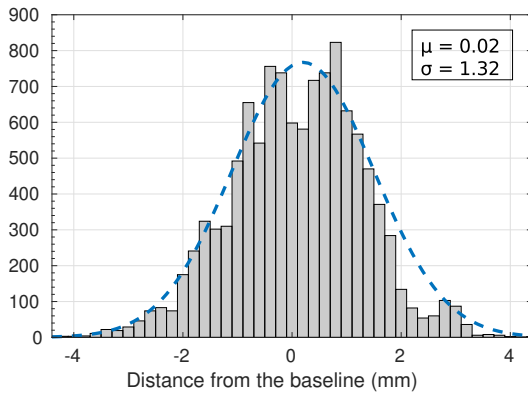


Fig. 11. The histogram of controlled fingertip locations. The mean is 0.02 and the standard deviation is 1.32.

B. Tracking Speed

One important indication of the system performance is the tracking speed by which the users can scan a text line but maintain the tracking accuracy. The analysis was conducted for the experiments where the users received the stimulation feedback. Fig. 12 shows the speeds for multiple repetitions of the experiment for the users.

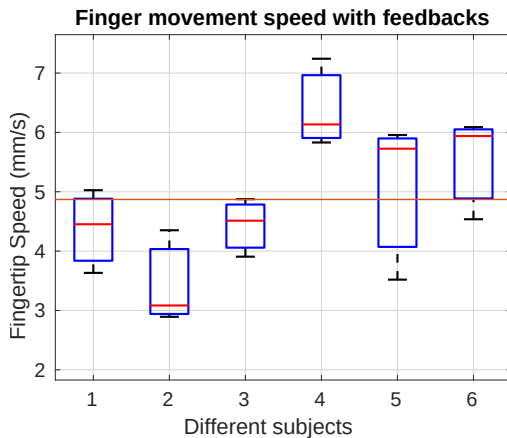


Fig. 12. Users' fingertip tracking speeds under multiple repetitions of the experiment with the feedback.

As shown in the figure, the average tracking speed is 4.87mm/s when the users receive the directional feedback commands from the system. One reason for this result, is that the users tend to stop their fingertip movement as soon as they receive a feedback stimulation command. We expect that the tracking speed under the feedback will be improved as the users become more accustomed to the feedback system.

In addition, Fig. 13 presents a comparison between the average tracking speeds for all users and all repetitions for receiving the feedback versus not receiving the feedback. A plot of the number and location of the up/down commands is added to the Fig. 13 to show the correlation between the number of feedback commands and the speed of the fingertip. It can be seen that in each instance, a drop in the speed of the fingertip is followed after a burst of up/down stimulation command. The results indicate that in the absence of the

feedback, users tend to start the tracking slowly, but their tracking speed increases as they get closer to the end of the text line. On the other hand, when users are receiving the feedback stimulation commands, they start very fast and then slow down as they progress. Our ongoing work is investigating this behavior for understanding the tracking control system that involves human in the feedback loop.

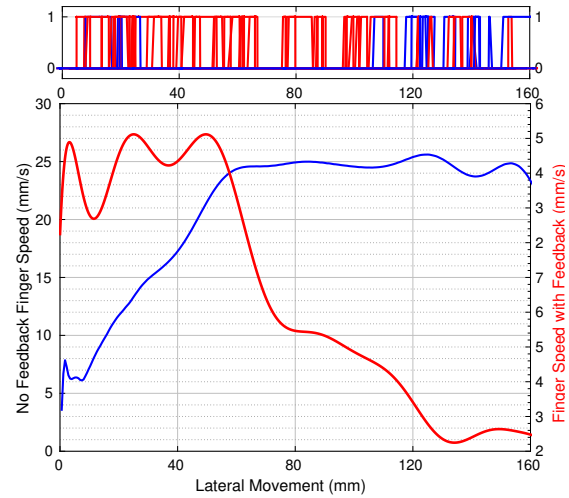


Fig. 13. Comparison between the average tracking speeds with and without the feedback: The blue line is the average speed of the natural finger movements in the absence of the feedback. The red line is the average speed of the finger movements that received the up and down stimulation feedback. Note that the scale of the Y-axis for the blue and red plots are different. The upper plot shows the up (blue) and down (red) electro-tactile stimulation commands that were sent to user subjects in all repetitions. Corresponding with the speed plot at the bottom, it shows that each time, a burst of commands is followed by a drop in the fingertip tracking speed.

V. CONCLUSION

In this work, based on a custom-built, finger-wearable and electro-tactile based refreshable Braille system, a new opto-electrotactile-feedback method that involves human-in-the-loop was proposed and developed to allow tracking and following the printed text lines by user's fingertip for "reading". The method incorporates electro-tactile stimulation with the developed rapid Optical Character Recognition (R-OCR) approach, which facilitates the new text-line tracking control strategy that enables text lines following by user fingertip during "reading". Extensive experiments were conducted to test the ability of blindfolded participants to follow the printed text lines using the opto-electrotactile-feedback method. The experimental results show that as the outcome of the feedback, the users were able to maintain their fingertip within a 2mm distance of the text while scanning a text line using the finger. The tracking speeds and the response times of the users in multiple repetitions were also examined and analyzed. It indicated that users with more experience with the system will take significantly less time to react to the commands. Our work is a significant step towards aiding the BVI users with a portable means to read any printed texts to Braille through the tracking and the translation in real time.

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